Akshat Sharma

519-992-7451| akshat@akshatsharma.ca | LinkedIn | GitHub | akshatsharma.ca

Experience

Software Developer Apr 2024 – Present
Tessonics *Windsor, ON*

- Eliminated hardware dependency for integration testing by building a **C++20** PLC simulator that mirrors real production traffic, enabling parallel CI runs and cutting integration test cycle time on the **RIWA** platform.
- Reduced end-to-end sensor latency to <10 ms by engineering a real-time Hall-effect current-detection algorithm in **C++20**, enabling sub-cycle fault detection on production lines.
- Delivered zero-downtime legacy integration by authoring backward-compatible 32-bit **C++20** DLLs exposing a drop-in API over **WebSocket**, bridging legacy clients to a modern **C++20** deep-learning inference server with no code rewrites.
- Cut manual release overhead to zero by building a **Python**-based **Debian** packaging workflow and self-hosted **GitHub Actions** runner, achieving continuous one-command deployment for all RIWA software.
- Owned end-to-end **PostgreSQL** schema design for the RIWA platform, authoring migration scripts and tooling to track schema evolution reliably across development, staging, and production environments.
- Accelerated frontend development by building **Go**-based **WebSocket** mock servers that replicate live hardware data streams, enabling UI load testing and iteration without physical device dependencies.
- Engineered production **STM32WB5MM** firmware for the SONYA sleep wearable with a full **BLE** GATT service enabling real-time wireless control, and built a self-testing binary that automated hardware validation across 50+ assembled boards, eliminating manual QC.

Machine Learning Research Assistant Jan 2023 – Apr 2024
Institute of Diagnostic Imaging Research *Windsor, ON*

- Explored and implemented VAEs, GANs, U-Nets, and Diffusion Models on custom ultrasound B-scan data to evaluate their viability for synthetic weld image generation in non-destructive evaluation.
- Collaborated on deep learning model development in **TensorFlow**; curated datasets, and developed Python scripts (using **OpenCV**, **NumPy**, **matplotlib**) for ultrasound image analysis in resistance spot weld evaluation.

Technical Skills

Languages: C | C++17/20/23 | Python | Java | Kotlin | Go | JavaScript

Data Science & ML: TensorFlow | PyTorch | Keras | OpenCV | NumPy | SciKit-Learn | matplotlib

Frameworks & Tools: React | Git | Docker | GitHub Actions | CMake | Conan

Database: PostgreSQL | SQLite | MySQL | Redis

Embedded / HW: I^2C | SPI

Projects

Charty | *Rust, ratatui, Yahoo Finance API, Finnhub WebSocket* Jan 2026 - Present

- Fetched historical OHLC data across 5 timeframes from Yahoo Finance v8 and streamed live trade ticks from Finnhub via **WebSocket** with exponential backoff reconnection (5 attempts, 2 s–32 s).
- Decoupled the **WebSocket** producer from the UI via a **Tokio** `mpsc` channel, enabling fault-tolerant live streaming with recoverable/fatal error discrimination and graceful unsubscribe on exit.
- Implemented a price alert system, watchlist, and market movers feed (top gainers, losers, most active) with pre/post-market session state detection.

Education

University of Windsor Jan 2021 – Apr 2024
Bachelor of Computer Science Honours *Windsor, ON*

- *Specialization in Artificial Intelligence, Minor in Mathematics*
- Relevant Coursework: Operating Systems, Concurrent Programming, Data Structures & Algorithms, Computer Networks, Database Systems